
資料庫系統

線上訂餐系統

G06

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| 應用情境

小明為了節省時間，在出門前先登入線上訂餐系統，瀏覽並分類查詢餐點，選擇想吃的餐點加入購物車。接著他確認訂單內容，系統自動進行訂單價格計算後，小明完成訂單建立。店家端收到訂單後，透過訂單管理功能開始製作餐點。當小明抵達店家時，餐點已製作完成，可以直接取餐，無需等待。

系統需求說明-功能性需求

顧客：

1.餐點瀏覽：

- 瀏覽餐點資訊
- 查看價格與介紹
- 搜尋餐點

2.訂單建立與管理：

- 建立、取消訂單
- 修改訂單內容

3.訂單歷史查詢：

- 查詢歷史訂單
- 查看訂單明細
- 查詢消費紀錄

4.訂單狀態通知：

- 接收訂單通知
- 查看處理進度

5.個人檔案管理：

- 註冊與登入
- 修改個人資料
- 變更密碼

6.付款管理：

- 選擇付款方式
- 完成線上付款
- 查看付款紀錄

系統需求說明-功能性需求

店家：

1. 菜單與餐點管理

- 新增、修改、刪除餐點品項與金額
- 管理餐點分類（注意限制條件：餐點必須關聯到某一個分類）
- 控制餐點的庫存狀態（如：上架、售完）

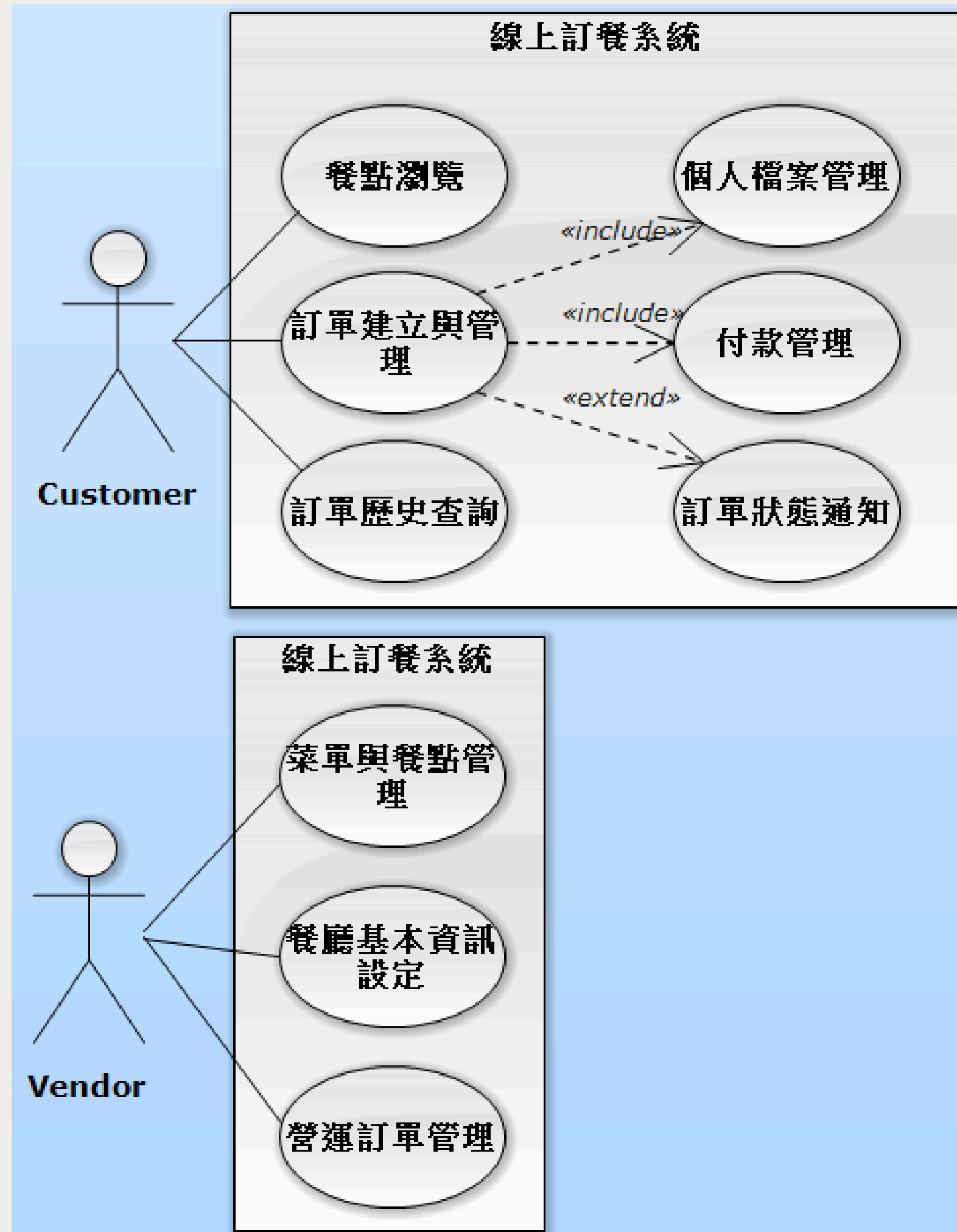
2. 營運訂單管理

- 接收並檢視顧客送出的新訂單
- 變更訂單狀態（接單、開始製作、完成）
- 查詢歷史訂單與營業額統計

3. 餐廳基本資訊設定

- 店家管理帳號設定，包含餐廳名稱、地址、電話等基本資訊
- 註冊與登入

使用案例



圖(一) 線上訂餐系統使用案例圖

完整性限制-實體完整性限制

- 用戶資料:用戶編號為主鍵，不得為空，且不可重複
- 菜單資料:菜單編號為主鍵，不得為空，且不可重複
- 商品資料:商品編號為主鍵，不得為空，且不可重複
- 訂單明細資料:明細編號為主鍵，不得為空，且不可重複
- 訂單資料:訂單編號為主鍵，不得為空，且不可重複
- 餐廳資料:餐廳編號為主鍵，不得為空，且不可重複

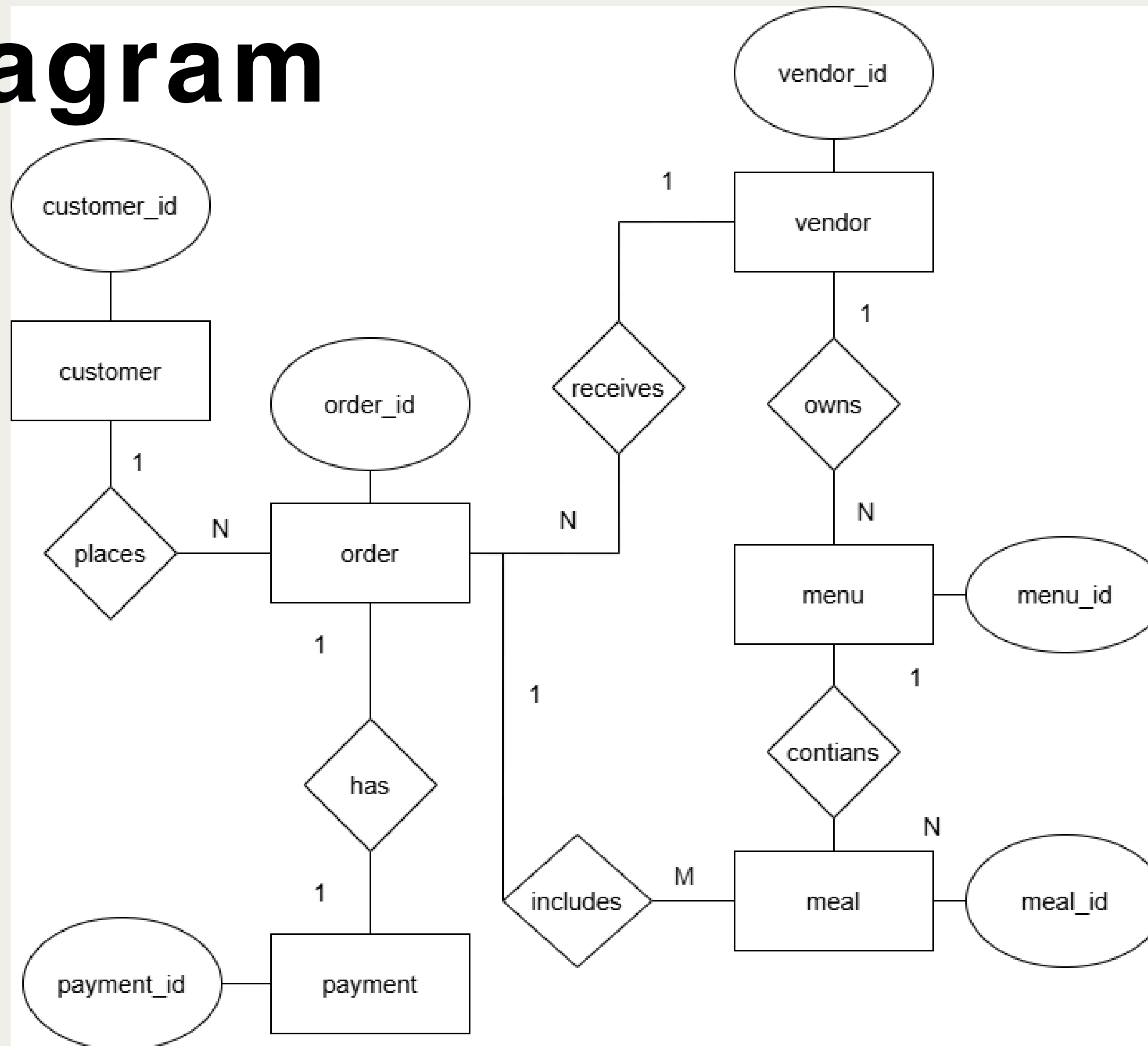
完整性限制-實體完整參考性約束

- 在訂單關係中，每筆訂單所對應的用戶，必須對應到用戶列表中存在的用戶
- 在訂單關係中，每筆訂單所對應的餐廳，必須對應到餐廳列表中存在的餐廳
- 在商品關係中，每個商品所對應的菜單，必須對應到菜單列表中存在的菜單
- 在菜單關係中，每個菜單所對應的餐廳，必須對應到餐廳列表中存在的餐廳
- 在訂單明細關係中，每筆明細所對應的商品，必須對應到商品列表中存在的商品
- 在訂單明細關係中，每筆明細所對應的訂單，必須對應到訂單列表中存在的訂單

| 完整性限制-欄位完整性限制

- 用戶ID、手機號碼、信箱、密碼不得為空
- 菜單內容不得為空
- 商品金額不得為空
- 庫存狀態不得為空
- 餐廳名稱、地址、電話不得為空
- 訂單編號不得重複
- 訂單日期、時間不得為空

ER Diagram



實體 -Customer

屬性	Domain	限制條件	說明
customer_id	INTEGER	PK	帳號唯一識別碼
customer_name	CHAR(15)	NOT NULL	顧客姓名
customer_phone	CHAR(10)	UNIQUE NOT NULL	電話號碼(聯絡或登入帳號)
class	ENUM('Customer', 'Vendor')	NOT NULL;DEFAULT 'Customer'	使用者的身分
password	VARCHAR(200)	NOT NULL	使用者密碼

實體 -Customer

The screenshot displays the MySQL Model (資料庫.mwb) interface. The left sidebar shows the database structure under 'mydb', including a 'customer' table with columns: customer_id, customer_name, customer_phone, class, and password. The main area shows a query window with the following SQL code:

```
1 INSERT INTO Customer
2 (customer_name, customer_phone, class, password)
3 VALUES
4 ('王小明', '0987654321', 'Customer', '654321');
5 SELECT * FROM Customer;
```

The query results are displayed in a table below the code:

	customer_id	customer_name	customer_phone	class	password
▶	1	傑哥	0912345678	Customer	123456
	2	王小明	0987654321	Customer	654321
•	NULL	NULL	NULL	NULL	NULL

實體 -Vendor

屬性	Domain	限制條件	說明
vendor_id	INTEGER	PK	帳號唯一識別碼
vendor_name	CHAR(15)	NOT NULL	店名
vendor_address	VARCHAR(50)	NOT NULL	店家地址
vendor_phone	CHAR(10)	NOT NULL	電話號碼(聯絡或登入帳號)
class	ENUM('Customer'、 'Vendor')	NOT NULL;DEFAULT 'Customer'	使用者的身分
password	VARCHAR(200)	NOT NULL	使用者密碼

實體 -Vendor

The screenshot shows a database management interface with a Navigator pane on the left and a main query editor on the right. The Navigator pane displays a tree structure of schemas, including 'mydb' which contains tables 'customer' and 'vendor', and other schemas like 'sakila', 'sys', and 'world'. The main query editor shows a SQL query for inserting a new vendor record into the 'Vendor' table. The query is as follows:

```
1 • INSERT INTO Vendor
2   (vendor_name, vendor_address, vendor_phone, class, password)
3   VALUES
4   (
5     '好難吃便當',
6     '雲林縣虎尾鎮文化路100號',
7     '0912345678',
8     'Vendor',
9     '123456'
10  );
11 • SELECT * FROM Vendor;
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has columns for 'vendor_id', 'vendor_name', 'vendor_address', 'vendor_phone', and 'class'. The first row shows the inserted record with 'vendor_id' 1 and the specified details. A second row shows a record with all NULL values.

	vendor_id	vendor_name	vendor_address	vendor_phone	class
▶	1	好難吃便當	雲林縣虎尾鎮文化路100號	0912345678	Vendor
•	NULL	NULL	NULL	NULL	NULL

The interface also includes a 'Form Editor' button on the right and a status bar at the bottom with 'Vendor 5' and 'Apply'/'Revert' buttons.

實體 -Menu

屬性名稱	Domain	限制條件	說明
menu_id	INTEGER	PK, Auto-increment	唯一識別碼
menu_vendor_id	INTEGER	FK	對應vendor.vendor_id
parent_id	INTEGER	FK, NULL	若為 NULL，代表它是「菜單」；若指向某個 ID，代表它是該菜單下的「分類」
menu_name	CHAR(10)	NOT NULL	菜單名稱

實體 -Menu

The screenshot shows the MySQL Modeler interface. The left sidebar displays the database structure for 'mydb', including tables 'customer', 'menu_category', and 'vendor'. The main window shows a query editor with the following SQL code:

```
3 VALUES
4 (1, NULL, '主餐');
5 • INSERT INTO Menu_Category
6 (vendor_id, parent_id, name)
7 VALUES
8 (1, 1, '便當類');
9 • INSERT INTO Menu_Category
10 (vendor_id, parent_id, name)
11 VALUES
12 (1, 1, '麵食類');
13 • SELECT * FROM Menu_Category;
```

Below the query editor, the 'Result Grid' is displayed, showing the data inserted into the 'Menu_Category' table:

	id	vendor_id	parent_id	name
▶	1	1	NULL	主餐
	2	1	1	便當類
	3	1	1	麵食類
•	NULL	NULL	NULL	NULL

The interface also includes a 'Form Editor' button on the right and 'Apply' and 'Revert' buttons at the bottom.

實體 -Meal

屬性名稱	Domain	限制條件	說明
meal_id	INTEGER	PK, Auto-increment	餐點唯一識別碼
meal_category_id	INTEGER	FK	關聯至 Menu.parent_id
meal_name	CHAR(20)	NOT NULL	餐點名稱
meal_description	VARCHAR(200)	-	餐點內容描述（如：包含青菜、滷蛋）
meal_price	DECIMAL(6,2)	NOT NULL, CHECK (>0)	餐點單價
meal_image_url	VARCHAR(200)	-	餐點照片的儲存路徑
meal_stock_quantity	INTEGER	DEFAULT 0, CHECK (>=0)	餐點剩餘份數
meal_available	ENUM('Available','Inactive')	DEFAULT 'Inactive'	上架狀態

實體 -Meal

The screenshot displays a database management interface. On the left, the 'Navigator' pane shows a tree view of databases including 'mydb', 'sakila', 'sys', and 'world'. Under 'mydb', the 'Tables' folder is expanded, listing 'customer', 'meal', 'menu_category', and 'vendor'. The main area shows 'Query 1' with an SQL INSERT statement for the 'Meal' table. The query includes column names and values for a new meal entry. Below the query editor, the 'Result Grid' is visible, showing the executed query's output with columns: id, category_id, name, description, price, and image_url. The first row contains the data for the inserted meal.

Query 1

```
1 • INSERT INTO Meal
2   (category_id, name, description, price,
3    image_url, stock_quantity, is_available)
4   VALUES
5   (
6     2,
7     '雞腿便當',
8     '包含青菜、滷蛋、白飯',
9     120.00,
10    '/images/chicken_bento.jpg',
11    50,
12    'True'
13  );
```

Result Grid

	id	category_id	name	description	price	image_ur
▶	1	2	雞腿便當	包含青菜、滷蛋、白飯	120.00	/images/d
•	NULL	NULL	NULL	NULL	NULL	NULL

實體 -Order

屬性名稱	Domain	限制條件	說明
order_id	INTEGER	PK, Auto-increment	訂單的唯一識別碼
customer_id	INTEGER	FK, NOT NULL	關聯至 Customer，標示下單的顧客
vendor_id	INTEGER	FK, NOT NULL	關聯至 Vendor，標示接單的店家
total_price	DECIMAL(10,2)	CHECK (>0)	訂單總金額
pickup_time	DATETIME	NOT NULL	顧客預定到店自取的時間
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	顧客下單的實際時間
order_payment_status	ENUM('Pending','Paid','Failed',)	FK	對應 payment.order_payment_statu

實體 -Order

The screenshot displays a database management interface. On the left, the 'Navigator' pane shows a tree view of schemas, with 'mydb' expanded to show tables: 'customer', 'meal', 'menu_category', 'orders', and 'vendor'. The main area is the 'Query 1' editor, showing an SQL insert statement for the 'Orders' table. The statement includes columns: 'customer_id', 'vendor_id', 'total_price', 'pickup_time', 'status', and 'payment_status'. The values provided are: 1, 1, 250.00, '2026-06-01 18:30:00', 'Pending', and 'Paid'. Below the query editor, the 'Result Grid' is visible, showing the results of the query. The grid has columns: 'order_id', 'customer_id', 'vendor_id', 'total_price', 'pickup_time', 'status', and 'payment_status'. The first row shows the inserted data: 1, 1, 1, 250.00, 2026-06-01 18:30:00, Pending, and Paid. The second row shows NULL values for all columns.

```
1 • INSERT INTO Orders
2   (customer_id, vendor_id, total_price,
3    pickup_time, status, payment_status)
4   VALUES
5   (
6     1,
7     1,
8     250.00,
9     '2026-06-01 18:30:00',
10    'Pending',
11    'Paid'
12  );
13 • SELECT * FROM Orders;
```

	order_id	customer_id	vendor_id	total_price	pickup_time	status	payment_status
▶	1	1	1	250.00	2026-06-01 18:30:00	Pending	Paid
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL

實體 -Payment

屬性名稱	Domain	限制條件	說明
payment_id	INTEGER	PK, Auto Increment	付款紀錄唯一識別碼
order_id	INTEGER	FK, NOT NULL	對應order.order_id
order_payment_status	ENUM('Pending','Paid','Failed',)	DEFAULT 'Pending'	訂單目前付款狀態
payment_method	ENUM('Cash','Credit Card','Line Pay')	NOT NULL	Cash、Credit Card、Line Pay
payment_time	DATETIME	NULL	付款完成時間

實體 -Payment

The screenshot displays a database management interface. On the left, the 'Navigator' pane shows a tree view of the 'mydb' schema, including tables like 'customer', 'meal', 'menu_category', 'orders', 'payment', and 'vendor'. The 'payment' table is selected. The main area shows a SQL query in the 'Query 1' tab:

```
1 • INSERT INTO Payment
2   (order_id, payment_status,
3    payment_method, payment_time)
4   VALUES
5   (
6     1,
7     'Paid',
8     'Line Pay',
9     '2026-06-01 17:45:00'
10  );
11 • SELECT * FROM Payment;
```

Below the query editor, the 'Result Grid' is visible, showing the results of the query. The grid has columns: 'payment_id', 'order_id', 'payment_status', 'payment_method', and 'payment_time'. The first row shows the inserted data: '1', '1', 'Paid', 'Line Pay', and '2026-06-01 17:45:00'. A second row with all 'NULL' values is also present.

	payment_id	order_id	payment_status	payment_method	payment_time
▶	1	1	Paid	Line Pay	2026-06-01 17:45:00
•	NULL	NULL	NULL	NULL	NULL

The interface also includes a 'Form Editor' button on the right and a 'Payment 9' tab at the bottom.

| 參考資料

[1] 虎尾野餐日線上訂餐網站 <https://shop.ichefpos.com/store/9iSDXw2e/ordering>

結束
感謝聆聽